

Solutions to Second Problem Sheet

on
Complex Analysis

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Topics: Metric Topology of $\mathbb{C}$
Limits and Continuity

“One cannot study with attention the functions of a complex variable without being struck by the harmony which reigns in this theory.”

-Augustin-Louis Cauchy

Problem 1

Discuss the convergence of the sequence $W = (w_n)$ where

a) $w_n = \frac{3n + 2i^n}{n},$

b) $w_n = \left(\frac{1+i}{\sqrt{2}}\right)^n,$

c) $w_n = \frac{n}{n+i}.$

In case of convergence, compute and sketch the first 7 terms.

Solution 1

a) Rewrite

$$w_n = 3 + 2\frac{i^n}{n},$$

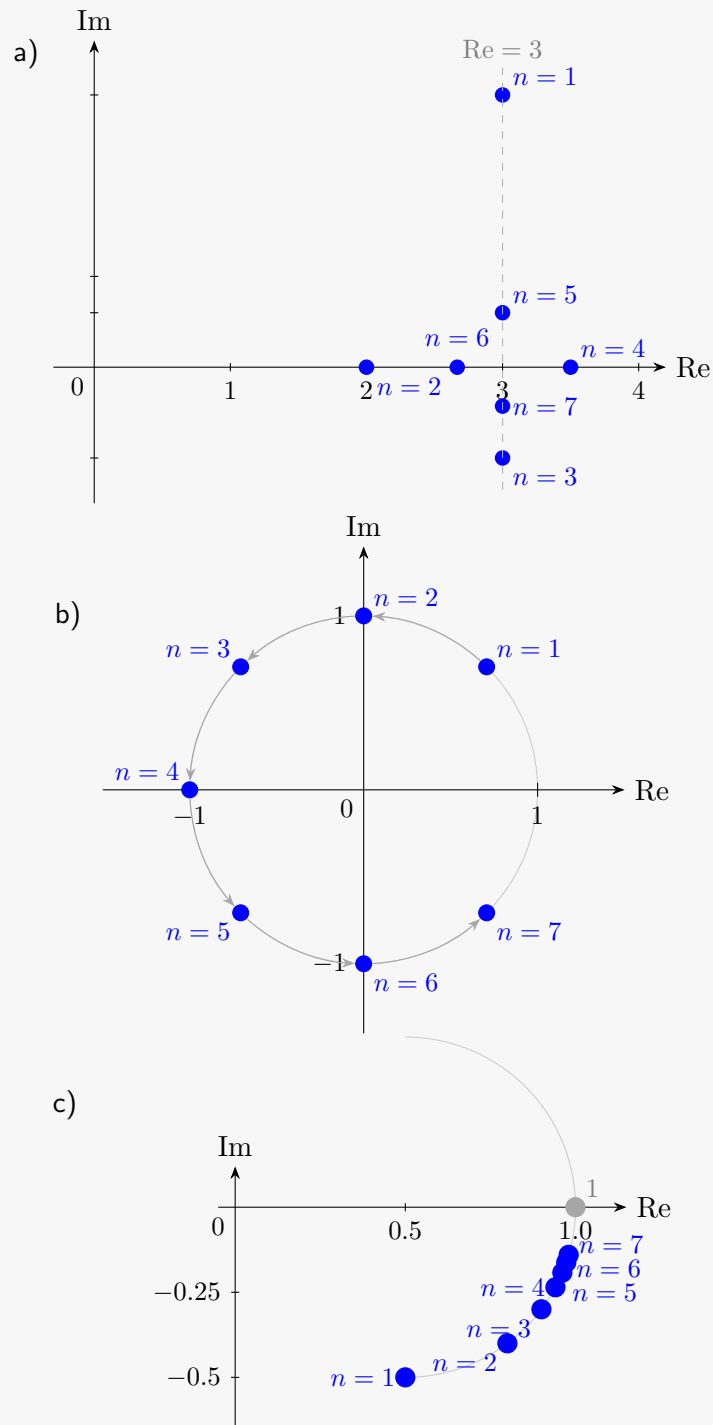
the first term clearly converges to 3 and the second one can be seen to converge to 0 by taking $N > 1/\varepsilon$ so that $|i^n/n| < \varepsilon$ provided that $n \geq N$. We conclude that the sequence converges to 3.

b) Since the term inside has modulus 1, the sequence oscillates in the unit circle around 0 (cf. Figure b) below).

c) Rewrite

$$w_n = \frac{1}{1 + \frac{i}{n}},$$

so that it can be shown that (w_n) converges to 1.



Problem 2

Show that the limit of a convergence sequence is unique.

Solution 2

Suppose that z, z' are two numbers satisfying the definition of convergence for a sequence, say (z_n) , then we get that given any $\varepsilon > 0$, there exist $K_1, K_2 \in \mathbb{N}$ such that

$$|z_n - z| < \frac{\varepsilon}{2} \quad \text{for } n \geq K_1,$$

and

$$|z_n - z'| < \frac{\varepsilon}{2} \quad \text{for } n \geq K_2,$$

and by the triangle inequality one can see that

$$|z - z'| \leq |z_n - z| + |z_n - z'| < \varepsilon \quad \text{for } n \geq K := \sup\{K_1, K_2\},$$

which means that $|z - z'| < \varepsilon$ for any $\varepsilon > 0$ and thus $z = z'$.



Problem 3

If $Z := (z_n)$ we define $\Re(Z), \Im(Z)$ to be the sequences $\Re(Z) := (\Re(z_n)), \Im(Z) := (\Im(z_n))$. Show that $\lim Z$ exists if and only if $\lim \Re(Z)$ and $\lim \Im(Z)$ both exist. In this case

$$\lim Z = \lim \Re(Z) + i \lim \Im(Z).$$

Solution 3

($\Rightarrow$) Notice that

$$|\Re(z_n) - \Re(z)| = |\Re(z_n - z)| \leq |z_n - z|, \quad |\Im(z_n) - \Im(z)| \leq |z_n - z|$$

so that if $(z_n) \rightarrow z$, then there exists $K \in \mathbb{N}$ such that $|z_n - z| < \varepsilon$ for all $n \geq K$ and the conclusion readily follows.

($\Leftarrow$) By the triangle inequality

$$|\Re(z_n) - L + i\Im(z_n) - M| \leq |\Re(z_n) - L| + |\Im(z_n) - M|$$

so that if both $\Re(z_n)$ and $\Im(z_n)$ converge to, say, some L, M , respectively, then we can bound $|\Re(z_n) - L| < \varepsilon/2$ and $|\Im(z_n) - M| < \varepsilon/2$, the conclusion is now immediate from the last inequality. In case of convergence we have

$$|z_n - z| = |\Re(z_n) - \Re(z) + i(\Im(z_n) - \Im(z))|$$

so that we take $L = \Re(z)$ and $M = \Im(z)$ and the equality of the limits now follows.



Problem 4

Suppose that every sequence in $\mathbb{R}$ converges if and only if it is a Cauchy sequence. Hence show that this result holds true for every complex sequence.

Solution 4

Let (z_n) be a Cauchy sequence, then given $\varepsilon > 0$ there exists $H \in \mathbb{N}$ such that $|z_n - z_m| < \varepsilon$ provided that $n, m \geq H$, with this in mind we see that $|\Re(z_n) - \Re(z_m)| < \varepsilon$ and $|\Im(z_n) - \Im(z_m)| < \varepsilon$ so that both the real and imaginary part of the sequence is Cauchy (and thus convergent) and hence the sequence must be convergent. The converse is easier, if the sequence is convergent, say to z , then given any $\varepsilon > 0$ there exists $H \in \mathbb{N}$ such that $|z_n - z| < \varepsilon/2$ and $|z_m - z| < \varepsilon/2$ provided that $n, m \geq H$, from this we conclude that (using the triangle inequality) $|z_n - z_m| < \varepsilon$.



Problem 5

Prove whether the set D is open or closed.

- a) $D = \overline{B}_\rho(z_0)$ with $z_0 \in \mathbb{C}$ and $\rho > 0$.
- b) $D = B_\rho(z_0) \setminus \overline{B}_r(z_0)$ with $0 < r < \rho$.
- c) $D = \{z \in \mathbb{C} : -\ell \leq \Re(z) \leq \ell \wedge -h \leq \Im(z) \leq h\}$ with $\ell > 0, h > 0$.

Solution 5

a) Is clearly closed, we show this by choosing $z \in \overline{B}_\rho^\sim(z_0)$ and $r = |z| - \rho > 0$ so that if $b \in B_r(z)$, then $|b - z| < r = |z| - \rho$ and hence $-|b| < -\rho$ which is equivalent to $|b| > \rho$ and this is just $b \in \overline{B}_\rho^\sim(z_0)$. We conclude that $B_r(z) \subset \overline{B}_\rho^\sim(z_0)$ and thus $\overline{B}_\rho^\sim(z_0)$ is open ($\overline{B}_\rho(z_0)$ is closed).

b) Clearly open, same argument as before (now directly with the set, not the complement) but now let $\gamma = \min(|z| - r, \rho - |z|)$ so that $B_\gamma(z) \subset B_\rho(z_0) \setminus \overline{B}_r(z_0)$ for any given $z \in B_\rho(z_0) \setminus \overline{B}_r(z_0)$.

c) This is a rectangle of 2ℓ by $2h$ centred at the origin. Given any point $z \in D^\sim$ we let $r = \min(|\Re(z)| - \ell, |\Im(z)| - h)$ so that $B_r(z) \subset D^\sim$ and thus D is open.

✧

Problem 6

Let $\rho > 0$, and let $z_0 \in \mathbb{C}$. Show that $B_\rho(z_0)$ is bounded and

$$\text{diam}(B_\rho(z_0)) = 2\rho.$$

Solution 6

2ρ is an upper bound since $|z - w| \leq |z - z_0| + |w - z_0| < 2\rho$ provided that $z, w \in B_\rho(z_0)$. Let $A = \{|z - w| : z, w \in B_\rho(z_0)\}$, then we show that $\sup A = 2\rho$. It is enough to verify that given $\varepsilon > 0$ we can find $a \in A$ such that $2\rho - \varepsilon < a \leq 2\rho$, we do this by letting $z = z_0 + \rho - \varepsilon/4, w = z_0 - \rho + \varepsilon/4$ and thus $a = |z - w|$ satisfies the required inequality since $a = |z_0 + \rho - \varepsilon/4 - z_0 + \rho - \varepsilon/4| = |2\rho - \varepsilon/2| = 2\rho - \varepsilon/2$ which is obviously greater than $2\rho - \varepsilon$ and the conclusion is now proved.

✧

Problem 7

Prove that

- a) $\lim_{z \rightarrow z_0} z^* = z_0^*$.
- b) $\lim_{z \rightarrow z_0} \alpha z^2 + \beta z + \gamma = \alpha z_0^2 + \beta z_0 + \gamma$.
- c) $\lim_{z \rightarrow \infty} \frac{1}{z} = 0$.

Solution 7

a) Trivial, use the fact that $|z^* - z_0^*| = |(z - z_0)^*| = |z - z_0|$.

b) Given $\varepsilon > 0$, choose

$$\delta = \min \left(1, \frac{\varepsilon}{2(|\alpha| + 1)(1 + 2|z_0|)}, \frac{\varepsilon}{2(|\beta| + 1)} \right),$$

so that it is straightforward to show that $|\alpha(z^2 - z_0^2) + \beta(z - z_0)| < \varepsilon$ provided that $|z - z_0| < \delta$.

c) Given $\varepsilon > 0$, choose $K = 1/\varepsilon$ so that $|z| > K$ implies $|1/z| < \varepsilon$.



Problem 8

Prove whether the following limits exist or not. If they exist, use the limit properties to justify it and if not, appeal to the divergence criteria.

a) $\lim_{z \rightarrow 0} \frac{z^*}{z}$

b) $\lim_{z \rightarrow 0} \frac{\Im(z)}{z}$

c) $\lim_{z \rightarrow i} \frac{z^2 + 1}{z - i}$

d) $\lim_{z \rightarrow 0} \frac{\Re(z)^2}{z}$

e) $\lim_{z \rightarrow 0} \frac{z}{|z|}$

Solution 8

a) Approach zero from the real axis so that the limit yields 1 and from the imaginary axis we get -1 so the limit does not exist.

b) Similar as before, approaching zero from the real axis we have that the limit is 0 (no number here has imaginary part) and approaching from the imaginary axis we have that the limit is 1 and thus the limit does not exist.

c) Rewrite the numerator as $(z + i)(z - i)$ so that the limit clearly exists and is $2i$.

d) The term can be bounded by $|z|$, so the limit exists and is 0 (apply the squeeze theorem for complex functions). e) Approaching zero from the positive real axis, we get that the limit is 1 and then from the negative real axis the limit yields -1 and thus the limit does not exist.



Problem 9

Let f, g be defined on $\Omega \subseteq \mathbb{C}$ to $\mathbb{C}$ and let z_0 be a cluster point of Ω . Show that if $\lim_{z \rightarrow z_0} f = 0$ and if g is bounded in some neighborhood of z_0 , then

$$\lim_{z \rightarrow z_0} fg = 0.$$

Solution 9

Given $\varepsilon > 0$, let $B_r(z_0)$ be such that $|g(B_r(z_0))| \leq M$ for some $M > 0$ and choose $\delta > 0$ such that $z \in B_\delta(z_0) \setminus \{z_0\}$ implies $f(z) \in B_{\varepsilon/M}(0)$. We conclude that for $z \in B_r(z_0) \cap B_\delta(z_0) \setminus \{z_0\}$ we have

$$|f(z)g(z)| \leq M|f(z)| < M \frac{\varepsilon}{M} = \varepsilon.$$



Problem 10

Let $\Omega \subseteq \mathbb{C}$, let $f : \Omega \rightarrow \mathbb{C}$ and let z_0 be a cluster point of Ω . If f has a limit w at z_0 and if $|f|$

denotes the function defined for $z \in \Omega$ by $|f|(z) := |f(z)|$, show that

$$\lim_{z \rightarrow z_0} |f| = |w|.$$

Solution 10

Easy, just use the fact that f has limit w at z_0 and the inequality $||f| - |w|| \leq |f - w|$.



Problem 11

Let f, g be defined on a subset Ω of $\mathbb{C}$ and let z_0 be an accumulation point of S . Show that

- i) If $\lim_{z \rightarrow z_0} f = 0$ and $|g| \leq |f|$ in a deleted neighbourhood of z_0 , then $\lim_{z \rightarrow z_0} g = 0$.
- ii) If $\lim_{z \rightarrow z_0} f = 0$ and g is bounded on some deleted neighbourhood of z_0 , then $\lim_{z \rightarrow z_0} fg = 0$.

This is sometimes known as the **squeeze theorem** (complex version).

Solution 11

i) Given $\varepsilon > 0$, choose $\delta > 0$ such that $|f(B_\delta(z_0))| < \varepsilon$ and thus let B be the intersection of $B_\delta(z_0)$ with the deleted neighbourhood of z_0 where $|g| \leq |f|$, so that

$$|g(B)| \leq |f(B)| < \varepsilon$$

and thus $g(z) \rightarrow 0$ as $z \rightarrow z_0$.

ii) Given $\varepsilon > 0$, choose $\delta > 0$ such that $|f(B_\delta(z_0))| < \varepsilon/M$ with M a bound for $|g|$ so that

$$|fg| \leq |f| |g| \leq |f| M < \frac{\varepsilon}{M} M = \varepsilon$$

on $B_\delta(z_0)$ and hence $(fg)(z) \rightarrow 0$ as $z \rightarrow z_0$.



Problem 12

Let f, g be defined on a subset Ω of $\mathbb{C}$ and let z_0 be an accumulation point of Ω . Show that if

$$\lim_{z \rightarrow z_0} f = L \quad \text{and} \quad \lim_{z \rightarrow z_0} g = M$$

with $M \neq 0$, then $\lim_{z \rightarrow z_0} \frac{f}{g} = \frac{L}{M}$.

Solution 12

It is enough to prove that

$$\lim_{z \rightarrow z_0} g = M \implies \lim_{z \rightarrow z_0} \frac{1}{g} = \frac{1}{M}$$

provided that $M \neq 0$, because of that is the case, then we can simply apply the theorem on multiplication of limits to get the desired result. To do so, assume that g has a limit M at z_0 , then for $\varepsilon = |M|/2 > 0$ ($M \neq 0$) we get that there is some δ' such that $|g(z) - M| < |M|/2$ for $z \in B_{\delta'}(z_0)$ which implies that $|M| - |g(z)| < |M|/2$ so $|g(z)| > |M|/2$ and hence $1/|g(z)| < 2/|M|$ for $z \in B_{\delta'}(z_0)$. Therefore, given any arbitrary $\varepsilon > 0$, let δ'' be such that $|g(z) - M| < \varepsilon|M|^2/2$

and choose $\delta = \inf\{\delta', \delta''\}$ so that

$$\left| \frac{1}{g(z)} - \frac{1}{M} \right| = \frac{|g(z) - M|}{|g(z)M|} \leq \frac{2}{|M|^2} |g(z) - M| < \varepsilon$$

for $z \in B_\delta(z_0)$ and the proof is complete. ✧

Problem 13

Assume that the function f does not vanish on a deleted neighbourhood of z_0 . Hence show that

$$\lim_{z \rightarrow z_0} f = 0 \iff \lim_{z \rightarrow z_0} \frac{1}{f} = \infty.$$

Solution 13

($\Rightarrow$) Suppose that $f(z) \rightarrow 0$ as $z \rightarrow z_0$, then given any $\alpha > 0$, we choose $\varepsilon = 1/\alpha > 0$ so that by assumption, there is some $\delta > 0$ such that $|f(z)| < \varepsilon$ for $|z - z_0| < \delta$ and thus $1/|f(z)| > \alpha$ for $|z - z_0| < \delta$.

($\Leftarrow$) Conversely, suppose that $1/f(z) \rightarrow \infty$ as $z \rightarrow z_0$, then given any $\varepsilon > 0$, choose $\alpha = 1/\varepsilon > 0$ so that by assumption there is $\delta > 0$ such that if $|z - z_0| < \delta$, then $|1/f(z)| > \alpha = 1/\varepsilon$ so that $|f(z)| < \varepsilon$ for $|z - z_0| < \delta$ and the proof is done. ✧

Problem 14

Discuss the continuity of the function

$$f = \begin{cases} \frac{|z|^3 - 1}{|z| - 1} z & \text{if } |z| \neq 1, \\ \lambda & \text{if } |z| = 1. \end{cases}$$

Find the values of λ for which the function is continuous at $1, -1, i, -i$. Can λ be defined such that f is continuous? Justify your answer.

Solution 14

Rewrite f as $(|z|^2 + |z| + 1)z$ for $|z| \neq 1$, and thus it is clear that the values for λ are $3, -3, 3i, -3i$, respectively so that λ cannot be defined in such way that f is continuous (it would require λ to simultaneously take the four given values). ✧

Problem 15

Show that if (f_n) is a sequence of continuous functions such that $f_n \rightrightarrows f$, then f is continuous.

Solution 15

Let the (uniform) convergence and continuity be on a set Ω and thus given $\varepsilon > 0$ we can find $K_1 \in \mathbb{N}$ such that $|f_n(z) - f(z)| < \varepsilon/3$ for $n \geq K_1$ and for all $z \in \Omega$, since each f_n is continuous, we can make $|f_k(z) - f_k(z_0)| < \varepsilon/3$ for any k provided that $z_0 \in \Omega$ and $|z - z_0| < \delta$ for some $\delta > 0$,

in particular, we choose $k \geq K$, and finally notice how

$$|f(z) - f(z_0)| \leq |f_k(z) - f(z)| + |f_k(z_0) - f(z_0)| + |f_k(z) - f_k(z_0)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

provided that $|z - z_0| < \delta$. We conclude that f is continuous on Ω .

